What Makes a Scientific Question Succeed?

Predicting Future Attention, Resolution, and Premise Revision from Question Structure and Evidence Context

Abstract

Large language models are increasingly asked to propose "important open questions", and are typically evaluated by asking another model how good those questions feel. We replace taste with history. We construct a temporally grounded dataset of **980 astronomy research questions** frozen at five historical cutoffs (2012–2020), spanning eight subfields and four distinct generation sources — evidence-tension mining, direct LLM elicitation, author-stated future work, and weak/negative controls — on top of the complete arXiv astro-ph metadata record (**313,189 papers, 2005–2025**). Each question carries a six-group cutoff-time feature record (text structure, evidence context, novelty, falsifiability, tractability, field environment) and a tiered outcome label derived from the five years of literature that actually followed. Under a strict, source-blinded outcome judge, 52.7% of questions were substantively addressed; negative-control questions were addressed least (32%) and direct-LLM questions most (88%), the latter carrying a documented parametric-leakage caveat. Interpretable models trained on 2012–2016 cutoffs predict 2020 outcomes at AUC 0.71 and transfer to entirely held-out subfields at AUC 0.65–0.71. Measured label noise implies a perfect-oracle ceiling of AUC 0.76 on these labels, so the model captures 83% of the attainable range above chance. Across univariate, controlled, ablation, and stability analyses, the robust predictors of future attention are **prior community recognition** (overlap with pre-cutoff review articles, $\beta = +0.34$, $p < 10^{-4}$), **the density of directly related prior evidence** ($\beta = +0.26$), and **explicitly stated competing hypotheses** ($\beta = +0.17$) — while high entity-specificity predicts *less* and *slower* engagement under literature-bounded labels. Five of six pre-stated mechanism hypotheses, including "evidence tension beats novelty", are *not* supported at scale — a result only visible because the dataset is large, multi-cutoff, and control-laden. Answering the structural-prior question posed in [23], we further show that within the tension-mined subset the generating evidence-graph structure itself is predictive: breadth of the paper cluster in tension predicts uptake ($\beta = +0.45$/SD), explicit contradiction predicts engagement, and quantified, cutoff-persistent tensions carry every resolution and premise refutation — a learnable, mining-time prior for structure-first question generation. Finally, a complete replication of the protocol in a maximally distant second domain — anti-aging skincare-ingredient research (19,193 PubMed abstracts, 1,043 backtested questions, a deterministic template generator and a checkable-condition judge) — reproduces the predictability (AUC 0.75–0.84), the control ordering, the breadth effect, the hypothesis failures, and the reliability ceiling ($\kappa = 0.22$ vs. 0.21), while the sign of specificity reverses exactly as the engagement-cost account predicts. All data, code, labels, and judge rationales for both domains are released.

1 Introduction

How scientific attention gets allocated is a central concern of the science of science, and it has been answered almost entirely at the level of papers and careers: which combinations of prior work earn citations [17], which research strategies scientists actually adopt [4], how impact is distributed across a field and a lifetime [3, 18]. The unit that precedes all of these — the research question itself, the bet placed before any paper exists — has no comparable empirical treatment.

The obstacle is not interest but measurement: questions leave no bibliographic record, so there is nothing to attach an outcome to.

This paper constructs those outcomes. We freeze research questions at historical cutoffs using only the literature available at the time, then read off what the following five years of literature did with each one. The resulting dataset makes a question-level analogue of the science-of-science program possible: rather than asking which papers were cited, we ask which *questions* the community took up, resolved, or overturned, and which of their cutoff-time properties predicted that.

The same construction answers a second, narrower question that motivated the series. Machine-generated research questions are currently evaluated by LLM or expert panels scoring "importance" and "novelty" at generation time — a measurement of how a question *sounds*, not of what it *does* [13]. The backtesting protocol of [23] introduced historical backtesting as the alternative: freeze questions using only pre-cutoff literature, then observe what the scientific community actually did afterwards. Its sealed v1.1 release has since grown from a ten-question pilot into a benchmark family — a scaled 424-question baseline instance, a four-cutoff temporal stress test (798 judged questions, 2010–2024), and a prospective instance frozen in 2026 — but its unit of comparison remains the generating *system*: it measures which generator families produce questions the future engages, not which measurable properties of an individual question predict that engagement. That predictor-discovery problem is this paper's subject.

This paper turns the backtesting protocol into a supervised-learning problem at scale. Three design decisions matter:

1. **The unit of data is (question, cutoff, subfield, source), not the question alone.** We generate questions at five rolling cutoffs (2012–2020) in eight astronomy subfields, so that one study contains what would otherwise be forty historical experiments, and era- or field-specific accidents can be detected rather than absorbed.

2. **No single generator.** Predictors learned from one generation system risk being that system's stylistic fingerprint. We use four sources — an evidence-tension mining system, a direct-LLM baseline, questions extracted from the authors' own future-work statements, and deliberately weak controls (random claim pairings, vague, untestable, and evidence-free-contrarian questions). Without negative controls a model can only learn that "question-shaped sentences succeed".

3. **Prediction is out-of-time or out-of-domain, always.** Random splits would place near-duplicate questions from the same era and topic on both sides. We train on 2012–2016 cutoffs, validate on 2018, test on 2020, and additionally hold out entire subfields.

The contribution is deliberately not a leaderboard number. It is a temporally grounded dataset of scientific questions with subsequent outcomes, and an analysis of *which properties of a question and its evidence context* predicted future scientific attention, resolution, and premise revision — with controls for field popularity, instrument eras, and generation method, and with stability checks across cutoffs, subfields, and sources.

2 Related work

2.1 Problem choice in the science of science

The choice of research problem is treated in this literature as a strategic act under uncertainty. Foster, Rzhetsky and Evans [4], analysing millions of MEDLINE abstracts, showed that scientists overwhelmingly pursue conservative strategies — deepening established chemical relationships rather than bridging distant ones — and that this conservatism is individually rational while collectively suboptimal, a finding later formalised as a question of how a field should allocate its collective search [12]. Uzzi et al. [17] reached a compatible conclusion from the citation side: the highest-impact papers combine a conventional core with a tail of novelty, so novelty by itself is not the operative virtue. Fortunato et al. [3] survey the broader program, and Wang and Barabási [18] give it book-length treatment.

Our results speak directly to this line. The predictor we find strongest — prior recognition of a question in review articles — is a question-level measurement of exactly the conservatism Foster et al. document at the level of research strategy, and our null result for semantic novelty is what Uzzi et al.'s conventionality finding would predict. The contribution is the unit of analysis: this literature observes what scientists *did* and infers the strategy, whereas we fix the candidate questions in advance and observe which were taken up. Because the questions are frozen before the outcome window opens, and because deliberately weak controls are included, the design separates properties of a question from properties of the field it sits in.

2.2 Generating and evaluating research questions

Machine generation of research questions and hypotheses has a long lineage, from literature-based discovery [16] to LLM systems that propose ideas or run entire studies [9, 19, 1], and it has been argued that question-asking is itself a core capability on the path to more general scientific intelligence [7]. Evaluation has not kept pace. The most rigorous instance of the dominant paradigm, Si et al.'s expert study [13], still measures opinion at generation time rather than what the ideas turned out to be worth. Outcome-based alternatives have appeared only recently and independently: HindSight [5] matches generated ideas against papers published after a temporal cutoff and finds that LLM-judged novelty correlates *negatively* with realised impact, and the ideation–execution study [14] had human researchers execute both LLM and human ideas, finding that the LLM ideas rated higher before execution scored lower after it. Both converge with our null results for novelty from different fields and different methods.

An earlier framework [22] built an evidence-graph system that generates questions from cross-paper observational tensions. The protocol paper [23] defined the backtesting benchmark and showed, in its pilot, that future literature substantively engaged all ten frozen exoplanet-atmosphere questions (one premise — a strongly subsolar water abundance for HD 209458 b — later refuted by three independent analyses, the exact convergence the question called for). Its sealed v1.1 release then scaled the evaluation and sharpened three findings this paper builds on. First, at $n = 125$ per baseline, engagement rates discriminate between generator families (random templates 73% vs. direct-LLM 96%, $p < 10^{-4}$), and random templates acquire a measurable 11%-answered floor — engagement metrics have a priced chance level. Second, a generator decomposition (LLM-only vs. deterministic evidence structure vs. structure-plus-LLM verbalization) crossed with a four-cutoff temporal stress test (2010–2024, 798 judged questions, the last window postdating the model's training) located the foresight signal in *pre-cutoff evidence structure* rather than model weights: LLM-only generation shows near-ceiling engagement and the closest phrasing to future literature at every cutoff, yet an answered rate indistinguishable from random templates and no premise refutations outside the deepest-history era. Third, a seven-rater agreement study (two blinded humans, five judge models, 90 items) found the two humans agree at only $\kappa = 0.17$ — indicting the outcome taxonomy, not any particular judge, and capping what absolute outcome rates can mean. Both papers evaluate question *generators*; neither learns question-level predictors, which is the question this paper addresses: *what, measurably, makes a question one the community will take up?*

2.3 Outcome labels and judge reliability

Scoring a strategy on held-out history is standard in quantitative finance, along with its documented failure mode — overfitting to the backtest itself [2] — which motivates the freezing rules we inherit; forecasting benchmarks apply the same logic to events resolving after training [21], and SWE-bench [6] showed how frozen data plus a fixed task format can reorganise a field around measurable progress.

Because our outcome labels are assigned by a language model, their reliability is itself a measured quantity rather than an assumption. Using LLMs as judges is now routine [20], and so is certifying them by agreement with other models — a practice the protocol paper's seven-rater study shows to be badly optimistic: on an analogous taxonomy two independent human annotators agreed at $\kappa = 0.17$, below every model–model pair. We therefore report agreement with the conventional interpretive benchmarks [8] attached, treat all absolute rates as rater-relative, and hold the judge fixed within every comparison.

To our knowledge no prior dataset attaches future-outcome labels to *questions* frozen at multiple historical cutoffs with controlled generation sources and negative controls.

3 Theoretical framework

Why should any property measurable at the cutoff predict what a scientific community subsequently does with a question? The empirical sections that follow are organised by a single account — *attention allocation as recognition-constrained foraging* — assembled from three premises, each with independent support in prior literature. We are explicit about its epistemic status at the end of the section: the account was articulated to organise this paper's results, not derived before them, and its falsifiable content is therefore prospective.

Premise 1: evaluation is the price of attention. A community's capacity to take up questions is bounded by researcher time, instrument access, and career risk. Before anyone invests in a question, someone must pay an evaluation cost: deciding whether it is well posed, whether it can be answered with accessible data, and whether an answer would be publishable. Bounded rationality [15] implies this cost is not paid exhaustively — it is avoided wherever a cheaper proxy exists.

Premise 2: prior recognition amortises the evaluation cost. Signals that the community has already evaluated a question — above all its appearance in a review article — let an individual researcher skip the evaluation and inherit its conclusion. This is cumulative advantage [10] operating at the level of questions rather than of people or papers: a recognised question gets re-recognised, because recognition is itself the cheapest available evidence of worth. Foster, Rzhetsky and Evans' finding that scientists overwhelmingly choose conservative, tradition-following research strategies [4] is the strategy-level face of the same economy.

Premise 3: researchers forage on cost-and-yield cues. Information-foraging theory [11] models an information seeker as following "scent" toward patches of high expected yield per unit cost. Read as foraging cues, the feature groups of Section 4.3 make differentiated predictions. A dense base of directly related prior evidence signals a rich patch in which existing tools, data, and comparison points apply, lowering the marginal cost of a contribution — whether or not that evidence is in tension. An explicitly articulated pair of competing hypotheses converts an open-ended evaluation problem into a designable discriminating study, lowering execution cost. Semantic novelty, by contrast, raises evaluation cost without lowering execution cost. And entity-specificity shrinks the set of researchers for whom the patch is cheap at all.

Together the premises order the predictors the paper tests:

(P1) prior recognition in reviews should be the strongest and most stable positive predictor of five-year attention;

(P2) density of directly related evidence should predict uptake independently of whether that evidence agrees — density, not tension;

(P3) explicit competing hypotheses should predict engagement and, conditional on engagement, resolution;

(P4) semantic novelty should be null to weakly positive at a five-year horizon — the question-level analogue of the finding that impact favours conventional cores [17];

(P5) entity-specificity should predict less and slower *measured* attention.

Two boundaries of the account deserve emphasis. First, it is a theory of where attention flows, not of what a question is worth; the discussion returns to the gap between the two, and the error analysis of Section 6.7 exhibits it concretely — the attention model's largest miss is a question the community engaged by overturning its premise. Second, the account does not cover the premise-refutation channel at all. Refutation is error-correction rather than attention-following: Section 6.6 finds it concentrated in quantified, recurrent contradictions — structures that persist across cutoffs until someone runs the decisive test. A foraging account predicts the crowd's path; it is silent about which accepted conclusion will fall.

Finally, the epistemic status. The six hypotheses of Section 5 were stated before the outcomes were known; this framework was assembled afterwards, as the interpretation the results select among prior theories — Merton's cumulative advantage, Foster et al.'s conservatism, Uzzi et al.'s conventionality, and the foraging model of information seeking. Its falsifiable content is prospective: transported to a new domain, it predicts that recognition and evidence density transfer as predictors while novelty remains null, and the frozen prospective instance of [23] puts that prediction on the record before the outcomes exist.

4 Dataset construction

4.1 Corpus

We harvested the complete arXiv astro-ph metadata record — titles, abstracts, submission dates, and categories — from 2005-01 through 2025-12 via the public arXiv API (**313,189 papers** after deduplication; `pipeline/harvest.py`). Papers are assigned to eight subfields by transparent weighted keyword rules over title+abstract (multi-membership allowed): exoplanet atmospheres (4,354 papers), protoplanetary disks (10,378), stellar activity (4,347), fast radio bursts (1,966), gravitational waves (17,989), galaxy evolution (21,390), cosmology tensions (16,403), and compact objects (34,266). The classifier is identical at every cutoff, so subfield corpora are time-sliced views of a fixed rule, not retrofitted topic models.

Each cutoff year $c \in \{2012, 2014, 2016, 2018, 2020\}$ defines a **past window** $[c-7,\, c]$ used for generation and features and a **future window** $(c,\, c+5]$ used only for labels. The five future windows all close by 2025-12-31, giving every cutoff the same 5-year outcome horizon.

4.2 Question generation (40 cells $\times$ quota 25)

For every (cutoff, subfield) cell:

- **Evidence-tension (10).** We mine the cell's past-window abstracts for tension statements (contradiction, discrepancy, unexplained-result markers), cluster them so that each cluster spans ≥ 2 distinct papers, and have an LLM phrase each cluster as one precise question grounded only in the mined sentences. Provenance (exact sentences, arXiv ids, dates) is stored with the question.

- **Direct LLM (5).** The baseline the field implicitly uses: the model sees a year-stratified sample of pre-cutoff titles and is asked for the most important unanswered questions, with an explicit knowledge-freeze instruction.
- **Future-work (5).** Sentences in which pre-cutoff authors themselves flag an open problem ("remains poorly constrained", "further observations are needed"), selected for topic diversity and converted faithfully into interrogative form. These are real scientists' questions.
- **Controls (5).** Two random claim-pairings across unrelated papers, one vague question, one untestable question, one evidence-free consensus challenge — all phrased by the same LLM so that surface style cannot separate controls from real candidates.

Near-duplicates within a cell are removed (TF–IDF cosine ≥ 0.75). Ten frozen evidence-graph questions from [22] join the dataset as a small additional source (cutoff 2020, exoplanet atmospheres), linking this study to the original system.

Realized totals: 980 questions — 374 evidence-tension, 200 direct-LLM, 196 future-work, 200 controls, 10 from [22]. Every non-FRB cell filled its full quota of 25; early fast-radio-burst cells are data-limited by history itself (the pre-2013 FRB literature contains 21 papers, yielding 11 questions at the 2012 cutoff), which we treat as a feature of honest backtesting rather than a defect. The dataset is regenerable bit-identically from the frozen LLM cache; rebuilding the corpus from scratch and regenerating produced byte-identical questions.

Temporal isolation is enforced by construction — every mined sentence and every title shown to a generator predates the cutoff — and verified by an automated validator over the frozen records (980/980 pass).

4.3 Feature record (six groups, all cutoff-time)

A. Text structure length, named entities, numbers, causal / comparative / mechanistic language, measurable quantities, explicit alternatives ("real or artifact"), falsification wording, yes/no form, and a composite specificity score.

B. Evidence context retrieval of the top-20 pre-cutoff subfield papers for each question: similarity mass (top-1, mean top-5, count above threshold), instrument diversity among retrieved papers, density of tension markers in the retrieved evidence, and the age profile of that evidence; plus native evidence-record features (paper count, year spread) for tension-sourced questions.

C. Novelty nearest-neighbour semantic distance to the pre-cutoff corpus, mean top-10 similarity, concept-pair novelty (do the question's two most distinctive terms co-occur in any pre-cutoff paper?), and overlap with pre-cutoff review articles (was the question already posed in reviews?).

D. Falsifiability / answerability rule-based markers plus a blinded LLM structured coding of {observable, competing hypotheses, quantitative test, falsification path} for every question.

E. Tractability facilities named in the question checked against a lexicon of $\sim$50 astronomical facilities with operational epochs: counts of already-operational vs. future instruments, archival-data markers, longitudinal / theory / large-sample requirements.

F. Field environment (controls) subfield publication volume and growth at the cutoff, review activity, share of all astro-ph output, and proximity of the subfield's next major facility launch (capped at 10 years). These are confounders to control for, not virtues of the question: without them a model mistakes hot fields for good questions.

4.4 Tiered outcome labels

Tier 1 (automatic candidates). For each question we retrieve future papers from its subfield's future window: relevant-paper count (cosine ≥ 0.18), weak-relevance count, first-follow-up date, lead time in months, and review recognition.

Tier 2 (strict, blinded LLM judge). For every question with any plausible future evidence (956 of 980; the rest are auto-labeled `not_addressed`), a GPT-4o judge sees the question and its top-8 retrieved future records — never the generation source. The rubric is deliberately strict and two-stage: the judge first classifies every candidate paper as **direct** (investigates this question's own objects, quantities, or claimed relationship), **adjacent** (same topic only), or unrelated, and only then assigns `addressed`, `answer_status` $\in$ {answered, partially_addressed, posed_but_open, premise_refuted, not_addressed}, `premise_status`, supporting ids, a rationale, and a confidence. `addressed` is *derived*: it requires at least one direct candidate. This rubric matters — a first-pass lenient judge (GPT-4o-mini, same evidence) called 94% of questions addressed by accepting topical overlap; the strict per-candidate rubric brings the rate to 52.7% and restores variance to every analysis downstream.

Tier 3 (independent verification). A second model (GPT-4o-mini) re-judges all premise_refuted cases plus a 20% random sample under the identical rubric ($n = 200$): raw agreement on `addressed` is 63.5% ($\kappa = 0.21$), status agreement 58.4% ($\kappa = 0.22$). The disagreement is one-directional — the weaker model is systematically more lenient, almost never the reverse (its "partially_addressed" precision against the strict judge is 0.99) — so we treat the strict labels as primary and release both, with all rationales and supporting ids, for audit. These κ values sit exactly at the measurement ceiling the sealed v1.1 release [23] established for outcome taxonomies of this kind: in its seven-rater study, two independent, careful human annotators agreed at only $\kappa = 0.17$, and every judge model matched a professional annotator about as well as the humans matched each other ($\kappa = 0.17$–0.26). We therefore adopt the protocol paper's reporting discipline: absolute rates are rater-relative, and every comparative claim in this paper is made under a single fixed judge.

4.5 Design of the retrieval and judging stages

The two information-access components deserve fuller specification.

Retrieval. Both the cutoff-time evidence features and the future-window label candidates use the same retriever: TF–IDF over title-plus-abstract with English stop-word removal, a 50,000-term vocabulary, and sublinear term-frequency scaling, fitted separately within each (subfield, window) corpus, with cosine ranking. Three properties motivate this deliberately classical choice over neural embedding retrieval. First, *temporal safety*: the cutoff-time retriever may see nothing newer than the cutoff, and a term-statistics model fitted on the past window satisfies this by construction, whereas any pretrained embedding model carries post-cutoff language in its weights — the retrieval-side twin of the parametric-leakage channel documented for generation. Second, *determinism and auditability*: term weights are inspectable, every ranking reproduces exactly from the frozen corpus manifest, and no model version enters the provenance chain. Third, a *conservative bias with a known direction*: term-based retrieval under-matches paraphrase, which pushes labels toward `not_addressed`; we prefer an engagement floor whose direction is known to an unquantified error, and the threshold-starvation curve of Section 6.7 makes the bound explicit. The cost — missed semantic matches, most severe for narrow object-level questions — is measured rather than hidden, via the imported evidence-graph questions of Section 6.7 and the specificity results.

Judging. For each question with any plausible future evidence (top-1 similarity above threshold; 956 of 980, the remaining 24 auto-labelled `not_addressed`), the judge receives the question, its cutoff date, and the top-8 retrieved future records — never the generation source. The rubric is two-stage per candidate before any outcome is assigned: each record is classified *direct* (investigates this question's own objects, quantities, or claimed relationship), *adjacent* (same topic only), or unrelated, with topical similarity explicitly insufficient for *direct*; `addressed` is then derived, requiring at least one direct candidate, after which the five-way status, premise status, supporting ids, a written rationale, and a confidence are recorded. All calls run at temperature 0 against a pinned model version and are cached, so the released labels regenerate exactly; rationales and supporting ids are released for audit. Throughout, the judge is treated as a measurement instrument with a measured error rate rather than as an oracle: its leniency direction, second-judge agreement, and the ceiling that agreement implies are quantified in Section 6.7.

5 Models and evaluation protocol

Baselines before models, interpretable models before ensembles: majority class; field-popularity-only; question-text-only (bag of words); then L2 and L1 logistic regression, random forest, and gradient boosting on the structured features; multinomial GBM for the five-way status; a Cox proportional-hazards model for time-to-first-follow-up; and within-cell ranking concordance between model scores and future paper counts.

All headline numbers use the **temporal split** (train 2012–2016, validate 2018, test 2020). Generalization is probed with **leave-one-subfield-out** and a fixed **cross-domain split** (train: exoplanet atmospheres, disks, stellar activity, galaxy evolution → test: FRBs, gravitational waves, cosmology tensions, compact objects). Random splits are not reported anywhere in this paper.

Predictor identification goes beyond feature importance: (i) univariate associations; (ii) controlled logistic effects with subfield, cutoff, source, and environment covariates; (iii) feature-group ablations; (iv) sign-stability of controlled effects across cutoffs, subfields, and generation sources. A predictor is called robust only if it survives all four.

Pre-stated hypotheses

H1 Cross-source evidence tension predicts future attention better than semantic novelty.

H2 Observation availability predicts short-term attention but not premise refutation.

H3 Explicit competing hypotheses raise the probability of being answered, given attention.

H4 Novelty shows an inverted-U: very high novelty reduces short-term attention.

H5 Field popularity inflates future paper counts without raising answer rates.

H6 High-tension $\times$ high-tractability questions succeed most.

6 Results

6.1 What happened to the questions

Of 980 frozen questions, **52.7% were substantively addressed** within five years of their cutoff: 474 partially addressed, 30 posed-but-open, 8 answered, 4 premise-refuted, 464 not addressed. Addressed questions drew a mean of 3.8 relevant future papers and were first engaged after a mean of 16.7 months. Rates are stable across cutoffs (0.45–0.57), confirming that no single era drives the results.

Generation source separates outcomes exactly as a functioning label should (Table 1, Figure 1).

Table 1: Addressed rate by generation source (strict blinded judge, 5-year horizon).

Source	Addressed
Direct LLM	0.88
Author future-work	0.51
Evidence-tension	0.47
Weak/negative controls	0.32
Evidence-graph [22] ($n = 10$)	0.20

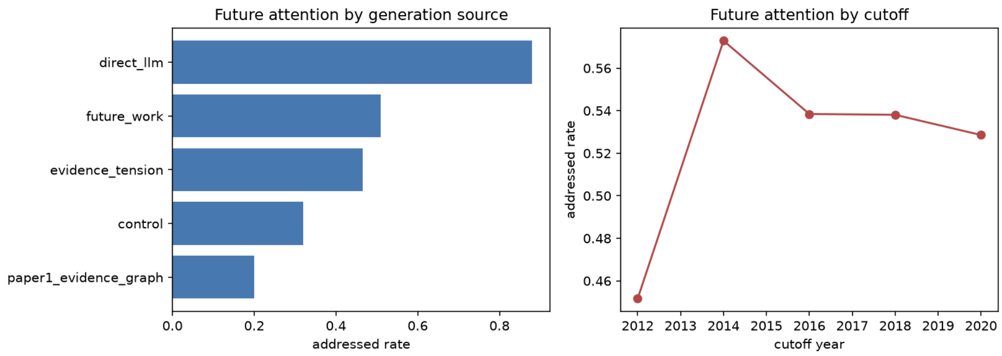

Figure 1: Addressed rates by generation source and cutoff.

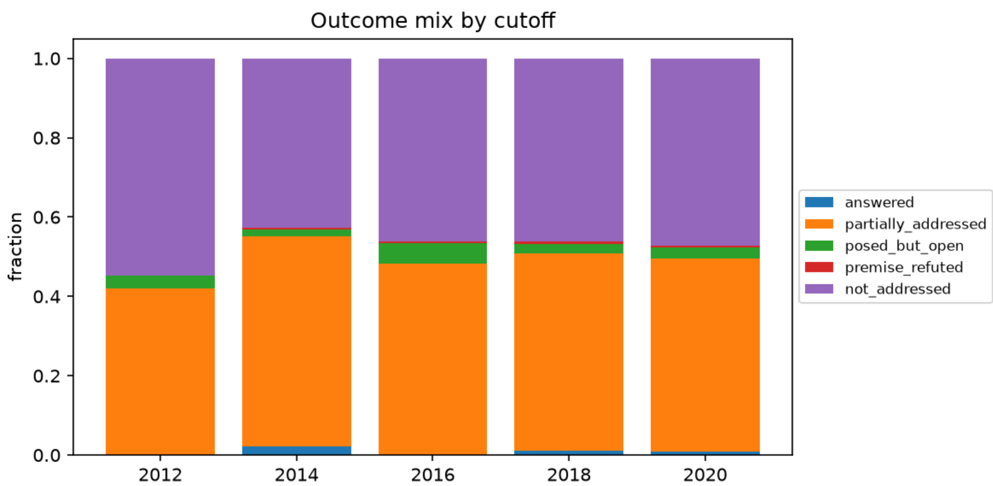

Figure 2: Outcome status mix across the 980 frozen questions.

Controls land lowest in aggregate — but the aggregate hides the dataset's most instructive split (Table 2). Random claim pairings, the purest chance-directed baseline, are addressed at 8.7%: that, not the pooled 32%, is the floor for chance-level engagement. Vague questions, by contrast, reach 0.850 — within three points of the direct-LLM rate and above every mined source — by being broad enough that some future paper always qualifies as direct. Coverage-style metrics are thus not merely gameable in principle; the worst-by-construction questions in the dataset game them in practice, which is why no engagement rate in this paper is interpreted without its source composition, and why the strict-bar analysis of Section 6.7 prices breadth explicitly.

Table 2: Outcomes within the negative-control arm, by subtype.

Control subtype	n	Addressed	Mean direct papers
Random claim pairing	80	0.087	0.17
Consensus challenge	40	0.275	0.45
Untestable	40	0.300	0.53
Vague	40	0.850	2.62

The direct-LLM rate of 0.88 should be read with care: the generating model may know, parametrically, which 2016-era questions became hot topics, an irreducible leakage channel we document rather than claim to eliminate (Section 8). the sealed v1.1 release [23] measures this channel directly and finds the same signature at scale: its direct-LLM baseline has the highest phrasing similarity to future literature of any system, near-ceiling engagement (96% at $n = 125$), yet an answered rate statistically indistinguishable from random templates and zero premise refutations outside the one era its weights could have memorized — "memorized relevance without specific foresight". The mined sources (tension, future-work) are built from pre-cutoff sentences and cannot leak this way; that they land at ∼0.5 against a strict judge is the honest base rate of real open problems. The ten evidence-graph questions [22] — all engaged by future literature under the protocol paper's adjudicated pilot evaluation (coverage 100%) — score only 2/10 here, a direct measurement of how much this paper's abstract-only retrieval and strict direct-engagement bar *under-count* engagement for narrow, object-level questions (they average 3.4 entities per question vs. 1.5 elsewhere). the protocol paper's own cross-instance anchor shows the same instrument-dependence from the other side: re-judging its ten questions on a 2.4× larger corpus flipped four labels in both directions — outcome rates are functions of the (corpus, retriever, judge) triple, and are comparable only within one instance.

6.2 Predicting future attention (Task A)

Temporal split, test = cutoff 2020 ($n = 210$, base rate 0.53); results in Table 3.

Table 3: Task A (future attention) on the temporal test split.

Model	AUC	Notes
Majority class	0.500	
Field popularity only	0.519	popularity is *not* a question-level predictor
Question text only (BoW)	0.772	absorbs source style + topic hotness
Logistic (structured, L2)	0.713	interpretable
Logistic (L1)	0.707	
Random forest	0.718	
Gradient boosting	0.688	

Three observations. First, everything question-level beats field popularity by a wide margin:

which questions succeed is not just which fields are hot. Second, raw question text is the strongest single signal — but it is also the least meaningful, because it can encode the generator's stylistic fingerprint (direct-LLM questions are both stylistically distinctive and 88% addressed). Third, the structured features reach AUC 0.71 while being *auditable*: every unit of signal is a named, cutoff-time property.

Out-of-domain. Trained on four subfields and tested on four entirely unseen ones ($n = 470$), the structured logistic model holds AUC **0.710** (GBM 0.647). Leave-one-subfield-out AUCs span 0.62–0.78 with every subfield above chance (stellar activity 0.78, gravitational waves 0.75, cosmology 0.71, disks 0.70, exoplanet atmospheres 0.69, galaxy evolution 0.66, FRBs 0.66, compact objects 0.62). Question-level predictors are not a memorized property of any one community.

Ablations. (GBM, temporal test; Figure 3): evidence-context is the load-bearing group — dropping it costs the most (0.688 → 0.665) and it alone nearly matches the full model (0.685). Text-only structure reaches 0.578, falsifiability 0.543, tractability 0.540, environment 0.550. Dropping the novelty group *helps* (0.708), foreshadowing the hypothesis results.

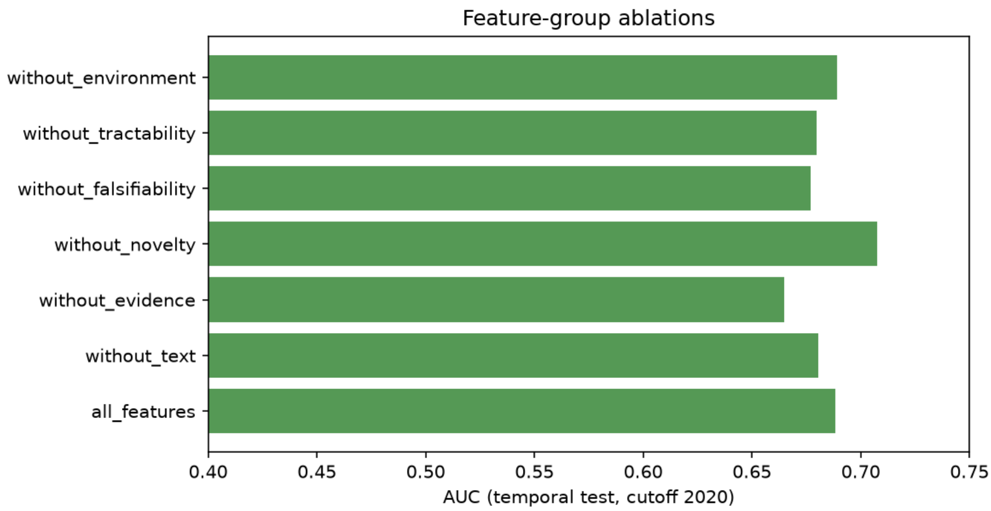

Figure 3: Feature-group ablations (GBM, temporal test split).

6.3 Which properties robustly predict attention

Controlled logistic effects (standardized, with subfield + cutoff + source + environment covariates), with sign-stability across the 5 cutoffs / 8 subfields / 4 sources, are given in Table 4 and Figure 4.

The picture that survives all four analyses:

1. **The community mostly answers questions it has already begun to recognize.** Overlap with pre-cutoff reviews is the strongest, most stable predictor — future attention is highly autocorrelated with present institutional attention, even after controlling for field size and growth.

2. **A dense base of directly related evidence predicts uptake; evidence *tension* per se does not.** What matters is that many papers already bear on the question, not that they disagree.

Table 4: Controlled effects on future attention with sign-stability fractions across cutoffs / subfields / sources.

Predictor	β	p	Sign-stable (cut./subf./src.)
Review overlap (already visible in pre-cutoff reviews)	$+\mathbf{0.34}$	4×10^{-5}	1.00 / 0.88 / 1.00
N. directly similar prior papers	$+\mathbf{0.26}$	0.005	0.80 / 1.00 / 0.75
Instrument diversity of prior evidence	$-\mathbf{0.21}$	0.005	1.00 / 0.75 / 1.00
Explicit competing hypotheses (LLM-coded)	$+\mathbf{0.17}$	0.021	0.80 / 0.88 / 0.50
Semantic novelty (distance to prior corpus)	$+0.17$	0.030	1.00 / 0.75 / 0.50
Entity count	-0.17	0.039	0.80 / 0.75 / 0.50
Specificity score	-0.16	0.041	0.80 / 1.00 / 0.50
Evidence tension density	$+0.06$	0.52	0.60 / 0.62 / 0.50

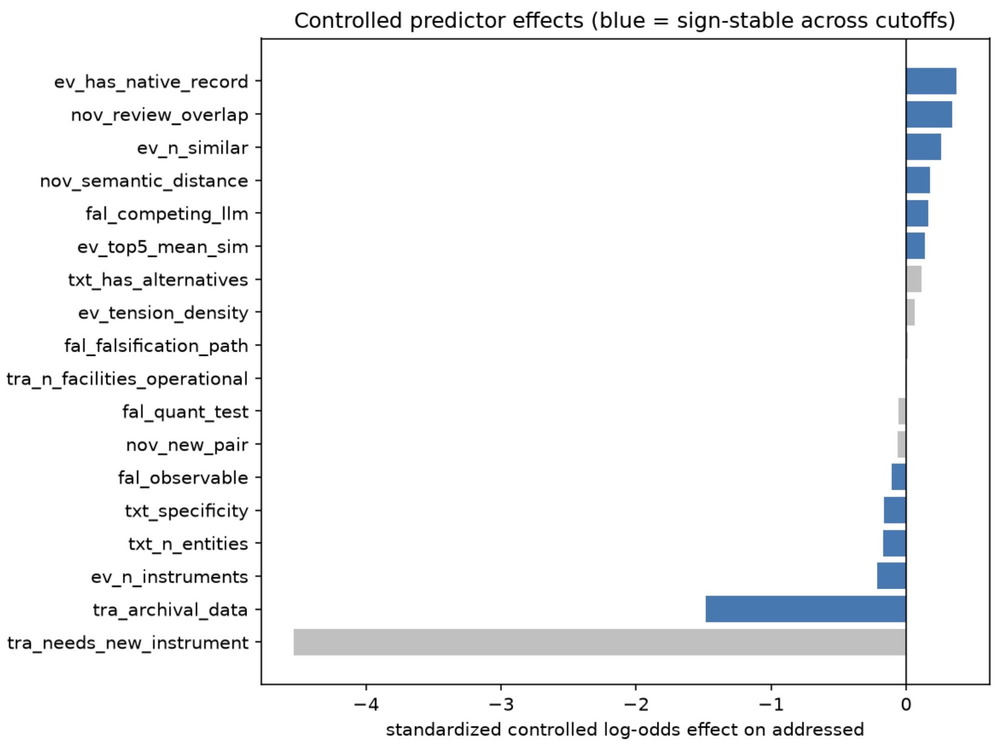

Figure 4: Controlled predictor effects on future attention.

3. **Questions whose evidence spans many instruments are *less* likely to be engaged** — consistent with multi-instrument questions being synthesis-level problems that no single group owns.

4. **Explicitly articulated competing hypotheses help** — the one falsifiability property with predictive teeth.

5. **Specificity cuts against measured attention.** Highly entity-specific questions are engaged less often and *more slowly* (Cox HR 0.73 per SD, $p < 10^{-4}$) under literature-bounded labels — partly a real narrowness effect, partly the retrieval bound (Section 8).

Time-to-engagement (Task C). In the Cox model (516 events/980), retrieval-similarity mass dominates speed (HR 3.77 per SD), field growth accelerates engagement (HR 1.15), an explicit falsification path accelerates it (HR 1.15), and specificity (HR 0.73) and tension density (HR 0.88) slow it. Within-cell ranking of 2020 questions by model score correlates with realized future paper counts at mean Spearman $\rho = 0.29$ across the 8 test cells.

Outcome type (Task B). The five-way status classifier reaches macro-F1 0.23 (accuracy 0.56): it separates addressed/partial from not-addressed but cannot yet learn the rare classes (8 answered, 4 refuted in total) — exactly the sample-size regime the design anticipated, and the reason Task A is primary at $n \approx 1{,}000$.

6.4 Pre-stated hypotheses: mostly falsified

Table 5: Verdicts on the six pre-stated hypotheses.

Hypothesis	Verdict
H1 tension > novelty	**No.** Novelty carries a small positive controlled effect ($\beta = 0.17$, $p = 0.03$); tension density is null ($\beta = 0.06$, $p = 0.52$).
H2 observation availability → attention	**No.** Operational-facility count is null on attention ($p = 0.91$); refutation side untestable (4 cases).
H3 competing hypotheses → answered	**Directionally yes, underpowered.** On attention: $\beta = 0.17$, $p = 0.02$. On answered-given-addressed: $\beta = 0.56$, $p = 0.08$.
H4 novelty inverted-U	**No.** Quadratic term null ($p = 0.46$).
H5 popularity → counts, not answers	**Untestable as stated.** Count effect positive but n.s. ($p = 0.21$); answer side degenerate (8 answered).
H6 tension × tractability interaction	**Marginal.** $\beta = 0.16$, $p = 0.053$ — suggestive, unconfirmed.

We regard this table as a feature of the method, not a failure of the paper: five of six mechanism intuitions that sound compelling in a proposal do not survive contact with a thousand historical outcomes and proper controls.

6.5 Interpretable prediction cards

The released pipeline emits, for any frozen question, a card of the form:

```
Question: How does the dark matter distribution in the innermost regions
of dwarf galaxies compare to the predictions of cold dark matter
simulations? [evidence_tension, cutoff 2020]
Predicted P(future attention): 0.50 → observed: addressed (partial)
Main positive factors: review overlap, directly similar prior papers,
explicit competing explanations
```

```
Main negative factors: high entity specificity, multi-instrument evidence
base
```

A cautionary card from the imported [22] questions: *"To what extent have actual JWST observations validated pre-launch predictions that aerosol-free TRAPPIST-1 CO_2 features should be detectable..."* (frozen at 2020-12-31, predicted 0.003, observed not addressed under our retrieval). The phrase "actual JWST observations" presumes data that did not exist at the cutoff — an instance of generation-time knowledge contaminating the *phrasing* of a nominally frozen question. Every LLM-mediated pipeline, including this one and that of [22], needs phrasing audits of exactly this kind — The sealed v1.1 release [23] released curation log institutionalizes them, recording presupposition and phrasing repairs made before freezing — and the dataset makes such audits possible because provenance is stored verbatim.

6.6 A learned prior over evidence-graph structures

The protocol paper [23] poses the question its architecture needs answered next: *which structural patterns of pre-cutoff evidence most often lead to questions the future answers, advances, or refutes?* Our dataset can answer it empirically, because every tension-mined question stores its generating evidence cluster verbatim — sentences, arXiv ids, dates. We derive a structural feature record for all 374 tension questions with the same transparent rules used elsewhere in the pipeline (`pipeline/structure.py`; frozen in `data/features/structure_features_v1.jsonl`): cluster breadth (papers and claims in tension), evidence-age profile, a four-way tension typology (explicit contradiction, unexplained result, methodological challenge, detection-vs-limit stance opposition), quantification of the discrepancy (N-sigma / factor-of markers), instrument structure, and **recurrence** — whether the same tension (shared source papers, same subfield) was already detectable at an earlier cutoff, a cutoff-time property since earlier past-windows are subsets of the current one. Table 6 and Figure 5 summarize the indicators.

Table 6: Evidence-graph structure indicators on the 374 tension questions. Controlled effects include subfield, cutoff, and environment covariates.

Structure indicator	Addressed rate	Controlled effect on addressed
4-paper cluster (vs. 2–3-paper)	0.54 vs. 0.32	$\beta = +\mathbf{0.45}$/SD, $p = 0.001$
Explicit contradiction markers	0.49 vs. 0.28	$\beta = +\mathbf{0.30}$, $p = 0.014$
Unexplained-result markers	0.49 vs. 0.43	$+0.19$, $p = 0.10$
Methodological challenge	0.46	-0.11, n.s.; 0/61 resolved
Quantified tension (N-σ)	0.48	null; carries all 3 refutations
Recurrent across cutoffs	0.49	null; carries all 6 resolutions

Three structural regularities emerge, one per outcome layer:

1. **Breadth buys attention.** The number of independent papers already in tension is the strongest structural predictor of both being addressed (0.54 for 4-paper clusters vs. 0.32 for 2–3-paper, Fisher $p = 6 \times 10^{-5}$) and future paper volume ($\beta = +0.12$, $p = 0.03$) — the structure-level counterpart of the question-level "density of directly related prior evidence" predictor of Section 6.3. Fresher evidence also draws more future papers (evidence age on volume: $\beta = -0.17$, $p = 0.002$).

2. **Explicit contradiction buys engagement; methodological doubt does not resolve.** Clusters whose sentences explicitly disagree are addressed at 0.49 vs. 0.28 without such markers ($p = 0.0095$), and five of the six resolved cases are contradiction-typed. Clusters built on methodological-challenge language engage the community at the average rate but resolved nothing (0/61) — consistent with the protocol paper's finding that questions must aim at a contestable claim, not at a diffuse worry about methods.

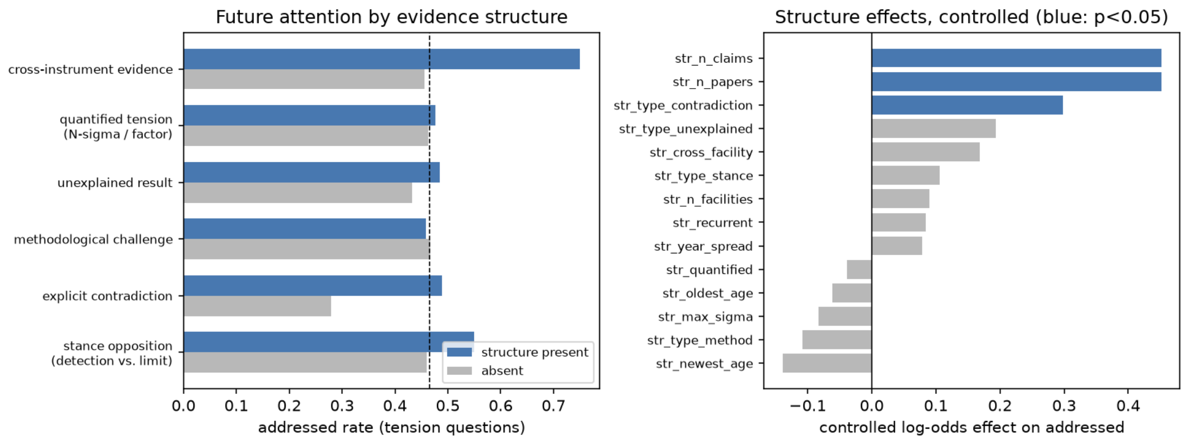

Figure 5: Evidence-graph structure and future attention. Left: addressed rates with each structure present vs. absent (dashed line: tension-question base rate). Right: controlled log-odds effects (blue: $p < 0.05$).

3. **Quantification and persistence mark the refutation channel.** All three premise refutations come from *quantified* clusters (3/63 vs. 0/311 unquantified, Fisher $p = 0.005$), and all six resolutions come from *recurrent* tensions (6/194 vs. 0/180 first-seen, $p = 0.03$) — structures the miner found again at two or more successive cutoffs. One honesty note bounds this: the three refutations trace to a single physical case (the FRB Galactic-latitude detection-rate discrepancy, independently re-mined at the 2016, 2018, and 2020 cutoffs and refuted by the same 2021 study), so they constitute one independent refutation event, and we report this layer as a case-level observation, not a statistical claim. Its shape is nonetheless exactly what the protocol paper's protocol predicts: a persistent, quantified, explicitly contradictory tension is a published conclusion waiting to be overturned.

The prior is learnable and transfers out of time: a logistic model on structural features alone, trained on the 2012–2018 cutoffs, ranks 2020 tension questions at AUC 0.586 ($n = 80$) — modest in absolute terms, but notable against the right comparison: restricted to the same single-source subset, the full 50-feature question-level record manages only AUC 0.502, because most of its cross-source signal is generator style and topic hotness that vanish within one source. Within a single generator, the generating *structure* is the recoverable signal. And because every structural feature is computable at mining time — before any question is phrased — the fitted coefficients (`results/structure_prior.json`) constitute a ranking prior for exactly the machine-scale structural search [23] proposes: enumerate candidate tensions, rank by breadth, explicit contradiction, quantification, and persistence, verbalize the survivors. The prospective instance (Section 9) is the pre-registered test of whether this prior holds on a future no model has seen.

6.7 Robustness, calibration, and the label-noise ceiling

Four analyses probe how much of the above depends on analytic choices, and what the measured label noise implies about the best any model could do (`results/robustness.json`; all computed from the frozen records).

The engagement bar. The released label derives `addressed` from at least one *direct* future candidate. Under stricter bars the levels drop, but the contrast that carries the leakage claim survives every bar, while the tension-vs-control contrast weakens into noise (Table 7). At $n_{\text{direct}} \geq 3$ tension questions sit nominally below controls (0.115 vs. 0.140), a crossover that is not significant ($p = 0.43$) and whose composition is diagnostic: 22 of the 28 controls clearing that bar are of the *vague* subtype. Vague questions accumulate "direct" hits because breadth makes the direct/adjacent distinction permeable, whereas engaged tension questions are engaged *thinly* — 1.9 direct papers on average, with 75% at one or two, against 3.2 and 42% for direct-LLM questions — and n_{direct} correlates negatively with entity count ($\rho = -0.12$) and specificity ($\rho = -0.17$, both $p < 10^{-3}$). The strict-bar crossover is thus the specificity effect and the breadth-farming failure mode meeting in one number, not a reversal of question quality.

Table 7: Addressed rate under stricter direct-engagement bars. Fisher p against controls.

Bar	All	Controls	Tension	Direct LLM	LLM vs. ctrl	Tension vs. ctrl
$n_{\text{direct}} \geq 1$	0.527	0.320	0.465	0.880	$p < 10^{-4}$	$p = 0.0007$
$n_{\text{direct}} \geq 2$	0.333	0.185	0.238	0.685	$p < 10^{-4}$	$p = 0.17$
$n_{\text{direct}} \geq 3$	0.212	0.140	0.115	0.515	$p < 10^{-4}$	$p = 0.43$

Retrieval bound and uncertainty. The stored top-8 similarity lists quantify how retrieval-bounded the labels are: at the released threshold ($\tau = 0.18$) 40% of questions already have no candidate above threshold, rising to 67% at $\tau = 0.22$ — the "lower bound" caveat in numbers. Bootstrap intervals (2,000 resamples) put the headline addressed rate at 0.527 [0.496, 0.557] and the non-control-minus-control gap at 0.259 [0.180, 0.330]. The judge's self-reported confidence is itself informative about its behaviour: it assigns full confidence chiefly when rejecting (addressed rate 0.235 at confidence 1.0 vs. 0.867 at 0.9), i.e. it is certain about absence and hedges about presence.

The label-noise ceiling. The independent second judge agrees with the primary on `addressed` 63.5% of the time. Modelling both as independent, class-symmetric noisy readers of a latent truth, that agreement implies a per-judge flip rate of 0.24 — and a *perfect* oracle for the latent truth would score only **AUC 0.758** against the primary labels. The observed temporal-split AUC of 0.713 is then **83% of the achievable range above chance**. The split of the measured disagreement between the judges is not identified — attributing more of it to the second judge raises the ceiling (to 0.90 if the primary judge's flip rate is 0.10) — but under any allocation the ceiling sits well below 1: raw AUC materially understates predictive skill on noisy-labelled outcomes, a point that applies to any study scoring open-ended outcomes with an imperfect judge.

Calibration and errors. Discrimination is not calibration: the temporal-split logistic model reaches Brier 0.224 with quintile-binned ECE 0.109, well calibrated at the extremes (top bin: predicted 0.87, observed 0.86) and overconfident in the middle (predicted 0.59, observed 0.45; Figure 6). Its false positives do not show inflated review overlap relative to true negatives (0.077 vs. 0.075), so the strongest predictor is not the failure mode. The most instructive error is a false negative: the model's second-lowest test-set prediction (0.089) is the HD 209458 b water-abundance question imported from [22] — precisely the question the community engaged by *refuting its premise*. The attention model assigns near zero to the dataset's clearest instance of a question doing what questions are for, which is the sharpest single demonstration that predicted attention and question value are different quantities (Section 3).

6.8 Cross-domain replication: anti-aging skincare ingredients

Every conclusion so far rests on one science. To test which of them are astronomy's, we ported the complete protocol — unchanged where possible, and changed only in directions this paper

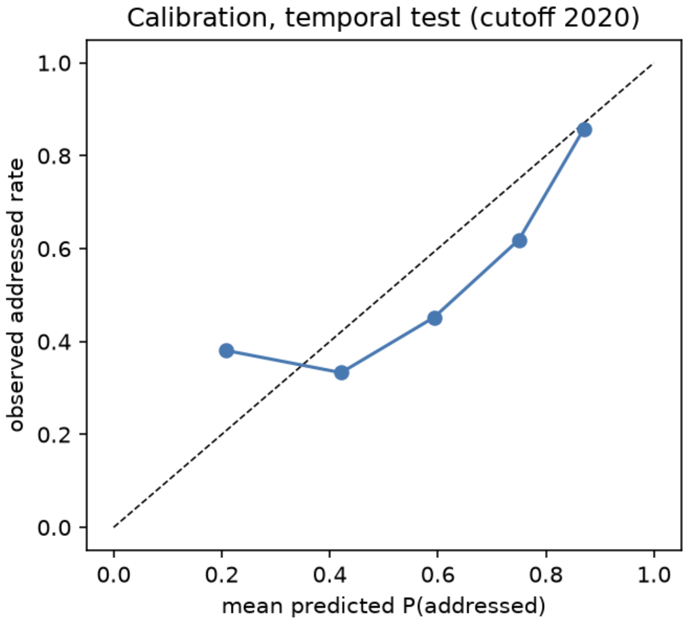

Figure 6: Reliability diagram, temporal test split (quintile bins, $n = 210$).

itself prescribes — to a maximally distant literature: anti-aging and anti-wrinkle skincare-ingredient research. The corpus is the complete PubMed record of English abstracts matching one reproducible skin-aging × ingredient query (19,193 papers, 2005–2026); the eight subfields become twelve ingredient classes; the facility lexicon becomes an assay-availability lexicon; and the same five cutoffs, past/future windows, source mix (tension, future-work, weak/negative controls), six feature groups, temporal and leave-one-class-out splits, pre-stated hypotheses, and structural prior are applied to **1,043 backtested questions**. Two deliberate deviations implement this paper's own v2 requirements (Section 8): questions are verbalised by deterministic templates over verbatim pre-cutoff sentences, removing the parametric-leakage channel entirely (no language model is called at any stage), and the outcome judge is the reliability-gated, checkable-condition rule our limitations section calls for — direct engagement requires retrieval relevance, an anchor-ingredient match, and a skin-aging outcome term; *answered* requires a direct randomised trial or meta-analysis — making every label reproducible bit-for-bit.

The protocol transfers (Table 8). Future attention is again predictable out-of-time (interpretable logistic AUC 0.746 on the 2020 test cutoff, $n = 226$; gradient boosting 0.836), again survives training on the six mature ingredient classes and testing on the six emerging ones (0.72–0.77), and again clears chance in every leave-one-class-out split (0.67–0.93). Controls again land lowest (26.8% vs. 40.3% for tension-mined questions), and the structural breadth effect of Section 6.6 replicates in sign and significance at about half the size ($\beta = +0.23$, $p = 0.04$). All six pre-stated hypotheses fail again: across the two domains, twelve pre-registered mechanism tests now yield one directional survivor, H3, marginal in both ($\beta = 0.56$, $p = 0.08$; $\beta = 0.71$, $p = 0.061$), while astronomy's marginal H6 interaction does not replicate ($p = 0.69$). Most tellingly, the reliability ceiling reappears under a different kind of judge: a blinded second judgement of a stratified sample gives 65.6% raw agreement and $\kappa = 0.22$ on *addressed* — against 63.5% and $\kappa = 0.21$ here — even though the replication's primary judge is a deterministic rule rather than a language model. The ceiling lives in the outcome taxonomy itself, not in any particular judge, which is what the noise-ceiling construction of Section 6.7 assumes.

What does *not* transfer is as informative as what does, because the changes go in the directions the theoretical framework predicts rather than in random ones. Exactly one predictor is sign-stable across every cutoff, class, and source *in both domains*: the density of directly

Table 8: The protocol and its findings across the two domains. Absolute rates and AUCs are comparable only within one column (the instances differ in judge and recalibrated retrieval thresholds).

	Astronomy (this paper)	Skincare replication
Corpus	313,189 arXiv astro-ph	19,193 PubMed
Backtested questions	980	1,043
Verbalization	LLM (temp. 0, cached)	deterministic templates
Outcome judge	blinded LLM rubric	checkable-condition rule
Addressed: controls / tension	0.32 / 0.47	0.27 / 0.40
Out-of-time AUC (interpretable / best)	0.713 / 0.718	0.746 / 0.836
Held-out-domain AUC	0.65–0.71	0.72–0.77
Evidence density β	+0.26	+0.79
Review overlap β	**+0.34**	0.00 ($p = 0.95$)
Specificity β (Cox HR/SD)	−0.16 (0.73)	**+0.57** (1.29)
Competing hypotheses β	+0.17	+0.28
Evidence tension density β	+0.06 (n.s.)	−0.09 (n.s.)
Hypotheses supported	0/6 (H3 marginal)	0/6 (H3 marginal)
Structural breadth β	+0.45, $p = 0.001$	+0.23, $p = 0.04$
Second-judge agreement (κ)	63.5% (0.21)	65.6% (0.22)

similar prior evidence ($\beta = +0.26$ here, $+0.79$ there) — the quantity Section 3 identifies with amortised evaluation cost, and the framework's central prediction. The recognition cue it rides on, however, is domain-specific: review overlap, astronomy's strongest predictor, is exactly null in the ingredient literature ($\beta = 0.00$, $p = 0.95$), displaced by naming a specific ingredient ($\beta = +0.76$) — in a field organised around named actives rather than named objects, the ingredient name *is* the recognition handle. (The effect is partly instrumental, since the rule judge anchors on ingredient names; the retrieval-bound caveat of Section 7 applies with its sign flipped.) Most strikingly, specificity reverses sign: negative and decelerating in astronomy ($\beta = -0.16$; Cox HR 0.73/SD), it is a strong positive predictor in skincare ($\beta = +0.57$, $p < 10^{-10}$) that *accelerates* engagement (HR 1.29, $p = 2 \times 10^{-4}$). The foraging account prices the reversal in engagement cost: a specific ingredient claim can be tested by one vehicle-controlled twelve-week split-face trial, so specific questions are cheap, high-yield patches; a specific astronomy claim needs scheduled telescope time that no group can allocate around one question. Where acting on a question is cheap, specificity is a yield cue; where it is expensive, specificity narrows the addressable community. A theory of attention allocation whose cue weights could *not* change across domains with different cost structures would be wrong about its own mechanism; this one changes them in the observed direction while keeping its central quantity fixed. Evidence tension, meanwhile, is null in both domains (+0.06 and −0.09): the H1 failure is not an astronomy accident.

Three caveats keep the comparison honest. The two instances differ in judge and in recalibrated retrieval thresholds, so absolute rates and AUCs are comparable only within one instance — the lesson of Section 4.5 applied to ourselves. The replication's refutation channel is empty (its two premise-challenged cases were both unconfirmed by the second judge and are treated as absent), so the refutation-side structural indicators could not be re-tested. And the rule judge's *answered* ("a direct trial or meta-analysis exists") is measurably more lenient than a human "settled": six of eight audited *answered* cases were downgraded to partial. The replication's full pipeline, frozen questions, features, labels, audit verdicts, and result tables are released alongside this paper's.

7 Discussion

What the dataset says about question value. The strongest honest summary: five-year scientific attention is predictable to a useful degree (AUC ∼0.7 out-of-time and out-of-domain) from cutoff-time properties, but the predictive properties are about the question's *relationship to its community* — already-in-reviews, dense direct evidence, articulated alternatives — more than about the intrinsic virtues the literature on "good questions" celebrates (novelty, tension, specificity). Attention follows preparation — the behaviour the recognition-constrained foraging account of Section 3 predicts and explains. Whether that is how science *should* allocate attention is precisely the kind of question this dataset now lets others study: the rare premise-refuted cases, the slow-burning specific questions, and the unaddressed-but-well-formed tail are all released with full provenance.

Specificity and the measurement bound. The negative specificity effect is partly real (narrow questions have small addressable communities) and partly instrumental: abstract-level TF–IDF retrieval misses engagement with object-level questions, as the evidence-graph subset demonstrates (2/10 here vs. 10/10 under the protocol paper's full-text evaluation). We report it as "less *measured* attention", never "worse questions" — and this is the main reason Task A labels should be read as lower bounds. the sealed v1.1 release [23] attacks the same confound from the metric side with a specificity-adjusted foresight score that down-weights broad questions anything would engage; under that rubric object-anchored questions are precisely what makes outcomes *resolvable* (its structure-first pipelines, 95–100% object-anchored, refute premises in every era; its LLM-only pipeline, 5% anchored, refutes almost none). The two results agree that raw engagement over-rewards breadth and under-rewards specificity; they differ only in the remedy — adjust the metric ([23]) versus control and caveat in the regression (here).

The direct-LLM anomaly is a warning, not a victory. The 88% addressed rate for direct-LLM questions is exactly what parametric leakage would produce: a model asked in hindsight for "the most important open questions of 2016" can simply recall what 2017–2021 worked on. Because source is a covariate in every regression, the predictor analysis is shielded from this; but any future benchmark that scores generators by addressed-rate alone will be gamed by hindsight knowledge. Mined sources with verbatim pre-cutoff provenance are the defensible foundation. The sealed v1.1 release [23] generator decomposition and temporal stress test now make this point causally rather than suggestively: over identical pre-cutoff evidence structures, a weight-free structural generator finds engaged questions in every era, the LLM adds value only as a separable verbalization layer, and the LLM-only pipeline's performance never rested on specifics that memorization could supply — its answered rate is flat across the model's training boundary because a topic prior needs no memory of the future. Our regression-level shielding (source as a covariate in every model) and the protocol paper's decomposition are complementary defenses against the same threat, operating at the analysis and generation levels respectively.

Implications for system design. The findings convert into three concrete practices. For *generator architectures*, they select an ordered pipeline: enumerate candidate evidence tensions at machine scale; rank them by the mining-time structure prior of Section 6.6 — cluster breadth, explicit contradiction, quantification, cross-cutoff persistence — and verbalise only the survivors, since phrasing adds resolution but no foresight [23]. Every ranking feature is computable before a question is written, so the prior adds no generation-time cost. For *evaluation design*, three practices transfer to any engagement-scored benchmark: include deliberately weak controls so that chance has a measured price (here pure chance costs 8.7%, not zero); price vagueness explicitly, because the worst-by-construction questions game coverage in practice and not only in principle (Table 2); and publish judged metrics only alongside measured reliability and the noise ceiling it implies, without which an AUC or an addressed rate cannot be interpreted. For *recommendation*, the two strongest predictors — prior recognition and evidence density — are computable from a live corpus, so a research assistant could rank candidate questions by predicted uptake today. That capability carries a Goodhart warning the framework makes precise: predicted attention as an optimisation target would amplify exactly the conservatism it measures, recommending the already-recognised and making recognition self-fulfilling. A deployment that cares about value rather than popularity should pair the attention model with the refutation-channel indicators — quantified, persistent contradictions — which point at overturnable conclusions rather than at the crowd's path.

8 Limitations

- **Abstract-level corpus.** Full texts are not used; tension and future-work mining see only abstracts, and citation-weighted attention is unavailable without a second data provider. Paper counts, lead times, and review recognition stand in for citation impact.
- **Retrieval-bounded labels.** A question can be engaged by literature our TF–IDF retrieval misses; not_addressed is a lower bound on engagement, uniformly across sources but not uniformly across specificity (see Section 7).
- **LLM judges, not humans.** Label reliability is estimated with an independent second model rather than human adjudication (63.5% raw agreement, $\kappa = 0.21$, disagreement one-directionally lenient). The sealed v1.1 release [23] seven-rater study reframes what such numbers can mean: on the analogous taxonomy, two independent human annotators agreed with each other at only $\kappa = 0.17$, every judge model matched the professional annotator as well as or better than the humans matched each other, and model–model agreement (κ up to 0.60) would have overstated reliability threefold. The weakness lives in outcome taxonomies of this kind, not in our judge specifically; accordingly, absolute rates here are rater-relative, all comparative results hold the judge fixed, and a reliability-gated taxonomy (fewer labels, checkable conditions, a measured human–human κ before release) is the shared v2 requirement for both papers. The annotation protocol, rationales, and supporting ids are released so human passes can replace or audit the judges; per-class precision tables are in the released agreement report.
- **Parametric-knowledge leakage.** The direct-LLM source (and, weakly, LLM phrasing of mined material) can import post-cutoff knowledge despite freeze instructions — including into question phrasing, as the JWST card shows. Source is controlled for in all analyses; the direct-LLM addressed rate should not be read as generator quality.
- **Rare outcomes remain rare.** 8 answered and 4 premise-refuted questions cannot support the Task B taxonomy or H2/H5 as stated; the design document's projection that these classes need 2,000–5,000 questions is confirmed empirically.
- **The noise ceiling is model-based.** The AUC ceiling of Section 6.7 assumes class-symmetric, independent judge errors, and neither assumption is innocent. Both judges read the same retrieved candidates, so their errors are plausibly positively correlated; for a fixed observed agreement that implies *more* per-judge noise and a *lower* true ceiling than reported, making the 83% skill fraction conservative. Asymmetric class errors cut the other way. The scan over error splits bounds the leverage of the symmetric assumption, but only human adjudication could replace the model with a measurement.
- **Two domains, one primary.** The full analysis is astronomy's; the skincare-ingredient replication (Section 6.8) shows the protocol and the evidence-density predictor transfer,

but it uses a different judge, its refutation channel is empty, and two domains do not span science. Fields whose engagement-cost structure differs from both — mathematics, say, where acting on a question is neither a trial nor an observation — remain open.

9 Conclusion

We built the first temporally grounded, multi-cutoff, multi-source dataset of scientific questions with future-outcome labels, and used it to replace two common practices: scoring questions by how they sound, and validating question generators on single-digit case studies. At $n = 980$ with strict blinded labeling, future scientific attention is predictable (AUC ≈ 0.71 out-of-time and out-of-domain, and 0.75–0.84 in an independent second-domain replication) — and the robust predictors are community-relational, not rhetorical. That number is 83% of the ceiling measured label noise permits, and the predictors are exactly those a recognition-constrained foraging account of attention allocation selects (Section 3); the ceiling construction itself transfers to any study that scores open-ended outcomes with an imperfect judge. Most pre-registered mechanism hypotheses failed, which is the point: a dataset an order of magnitude larger than its predecessor makes intuitions testable, and most did not survive. The dataset, every prompt, every judge rationale, and the full pipeline are released for the community to reuse, audit, and extend to other sciences — and the skincare-ingredient replication of Section 6.8 shows that the extension is a port, not a rebuild: the protocol, the evidence-density predictor, the hypothesis failures, and the reliability ceiling all reproduce in a second science, with the sign flips the engagement-cost account predicts. An extension to a third domain with harder outcome signals is in preparation (clinical oncology, where trial registrations and phase transitions are observable events rather than retrieval-bounded labels). A forward-looking test is already armed: the frozen prospective instance of [23] — 200 questions from four generators, frozen 2026-08-17 with a pre-registered 2027–2030 scoring window — will let the predictors learned here be evaluated against a future no model has seen, converting this paper's retrospective claims into ones that time itself will grade.

Reproducibility

All code, the frozen dataset, labels, and results are released; every pipeline stage is a single script, all LLM calls are cached and pinned to temperature 0, and the offline test suite plus a temporal-isolation validator run in CI. Rebuilding the corpus from the public arXiv API and regenerating the dataset reproduced the frozen questions byte-for-byte. The frozen dataset is additionally published on the Hugging Face Hub, loadable in one call, with the structure-prior and second-judge subsets as separate configurations. Repository and dataset URLs are withheld here for double-blind review and will be supplied on acceptance. The cross-domain replication of Section 6.8 is released as a separate repository with the same layout — code, frozen questions, features, labels, audit verdicts, and result tables — (URL likewise withheld for double-blind review); given its harvest, every replication number is deterministic (no language-model calls at any stage).

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) to draft and edit manuscript prose, to typeset the manuscript in LaTeX, and to assist in writing the analysis code for the evidence-structure prior (`pipeline/structure.py` and `scripts/run_structure_prior.py` in the released repository). After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Separately, and as a matter of method rather than of manuscript preparation, large language models are a component of the study itself: they phrase the mined questions and assign the outcome labels. Their role, model versions, prompts, and cached outputs are specified in Sections 3.2 and 3.4 and released in full with the dataset.

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